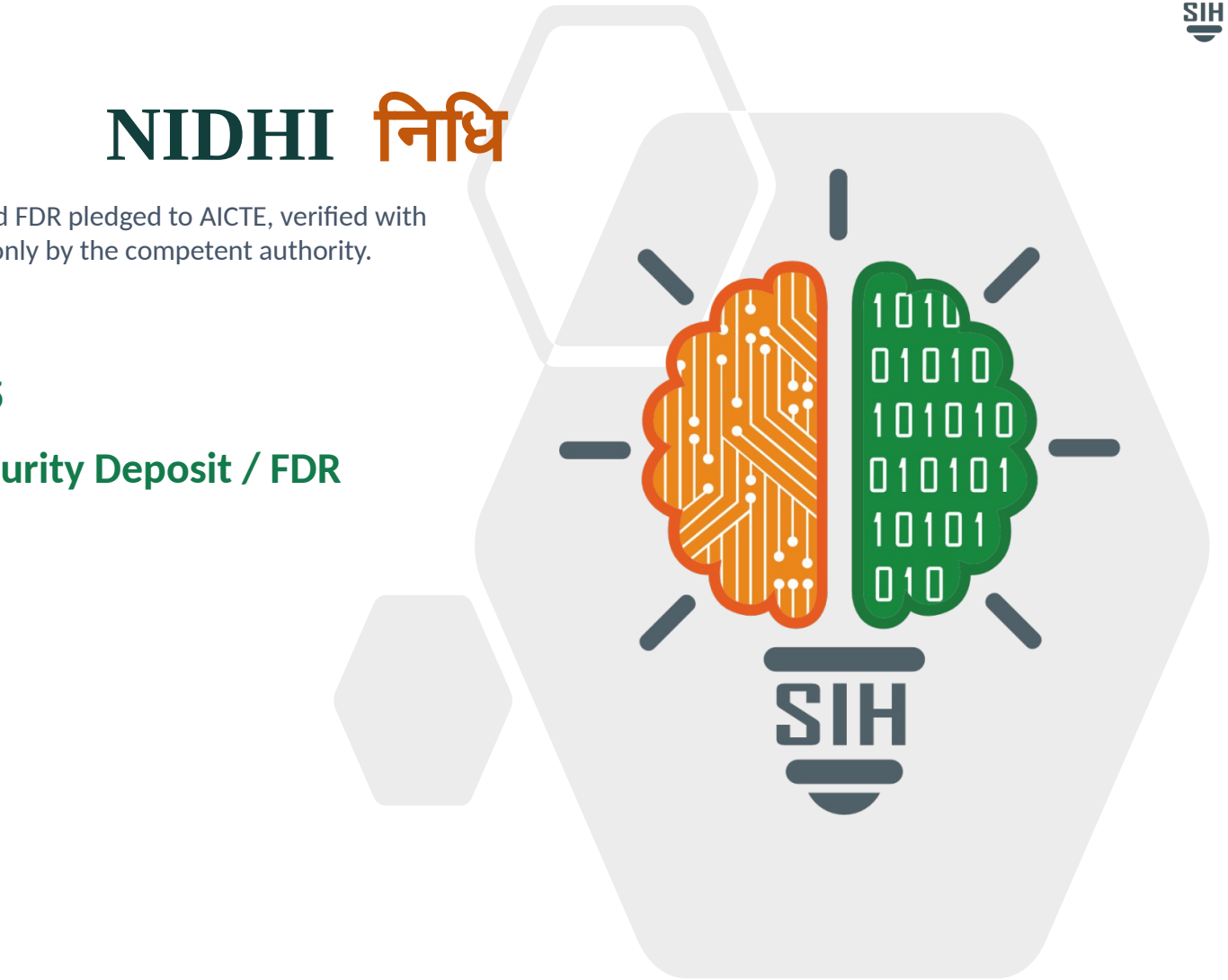
SMART INDIA HACKATHON 202

SMART INDIA HACKATHON 2026

NIDHI निधि

"A fund held in trust": every security deposit and FDR pledged to AICTE, verified with the bank, tracked to maturity and released only by the competent authority.



- **Problem Statement ID –** 26265
- **Problem Statement Title –** Security Deposit / FDR Management Portal
- **Theme –** Smart Automation
- **PS Category –** Software
- **Team ID –** 198001
- **Team Name –** SignalNine

SignalNine

NIDHI (निधि)

SMART INDIA HACKATHON 2026

One portal for every security deposit and FDR AICTE holds, from lodging to lien release

Problem today

- **Files, letters and spreadsheets.** Deposit status lives in many places, so no one has an up-to-date view of what is held and what is pending.
- **Paper is taken on trust.** An FDR copy can be altered, reused by a second institution, or closed at the branch after AICTE accepts it.
- **Maturity slips by.** Renewals are chased by hand; a lapsed FDR is security in name only.
- **Release by correspondence.** Conditions, approvals and the lien-release letter are tracked on file notings.

Proposed solution

- **Lodge online:** the institution uploads the FDR copy; the portal **reads it** and fills in the form.
- **Three sources side by side:** declared, on the document, confirmed by the bank. Never merged.
- **Monitor to maturity:** reminders at 90/60/30/7 days, escalation at 15 and 45 days after maturity.
- **Release by maker-checker** with a lien-release letter anyone can verify by QR.

Innovation and uniqueness

1. **Three sources, never mixed**
   Every field shown as declared, as read from the FDR copy, and as confirmed by the bank. A red cell is a difference.
2. **Reads the FDR copy**
   Reads PDFs, scans and phone photos (OCR, offline) and catches an altered copy on lodging (**100%** digital, **99.6%** scanned).
3. **Fraud signals**
   Same FDR reused by two institutions, bank with no record, FDR later closed at the bank (periodic re-confirmation).
4. **Verifiable letters**
   Acknowledgements and lien releases carry a QR and a SHA-256 fingerprint; a branch checks them without signing in.
5. **Late-renewal prediction**
   Ranks held FDRs by the chance the replacement comes after maturity, with reasons, 60 days ahead.
6. **Ask in English or हिंदी**
   “FDRs maturing in 30 days” becomes the right filters. Runs on the server: **no cloud AI**.

All 10 required capabilities, built:

✔ Repository	✔ Documents	✔ Verification
✔ Maturity tracking	✔ Renewals	✔ Release workflow
✔ Alerts & escalation	✔ Dashboard & reports	✔ Audit & access
	✔ Integration & security	

Declared, on the document, confirmed by the bank
Kept apart and never overwritten. A red cell differs from what the institution declared.

	Declared by institution 24 Oct 2026	Read from FDR copy 100% of fields read	Confirmed by bank Online, from the bank · 25 Oct 2026
FDR number	TD-914158-26	TD-914158-26 ✓	TD-914158-26 ✓
Amount	₹5,00,000	₹50,000 ✗	₹5,00,000 ✓
Issue date	21 Oct 2026	21 Oct 2026 ✓	21 Oct 2026 ✓
Maturity date	21 Oct 2029	21 Oct 2029 ✓	21 Oct 2029 ✓
Depositor	Medhavi Education Society	Medhavi Education Society ✓	Medhavi Education Society ✓
Lien in favour of	Member Secretary, AICTE	Member Secretary, AICTE ✓	Member Secretary, AICTE ✓

The deposit file: the FDR copy reads ₹50,000 where ₹5,00,000 was declared; the bank confirms the declaration.

2 open discrepancy(ies) — Raise one
Checked automatically against the deposit norm, the document, the bank and every other FDR on record.

Critical **Same FDR already lodged with AICTE** rule

FDR VN24417735 of Vindhya is already lodged as NDH-2025-000221 by Tapasya College of Engineering, Vijayawada.

Expected a fresh FDR · Found NDH-2025-000221

NDH-2025-000221, Tapasya College of Engineering, Vijayawada

Fraud signal: the same FDR lodged by a second college

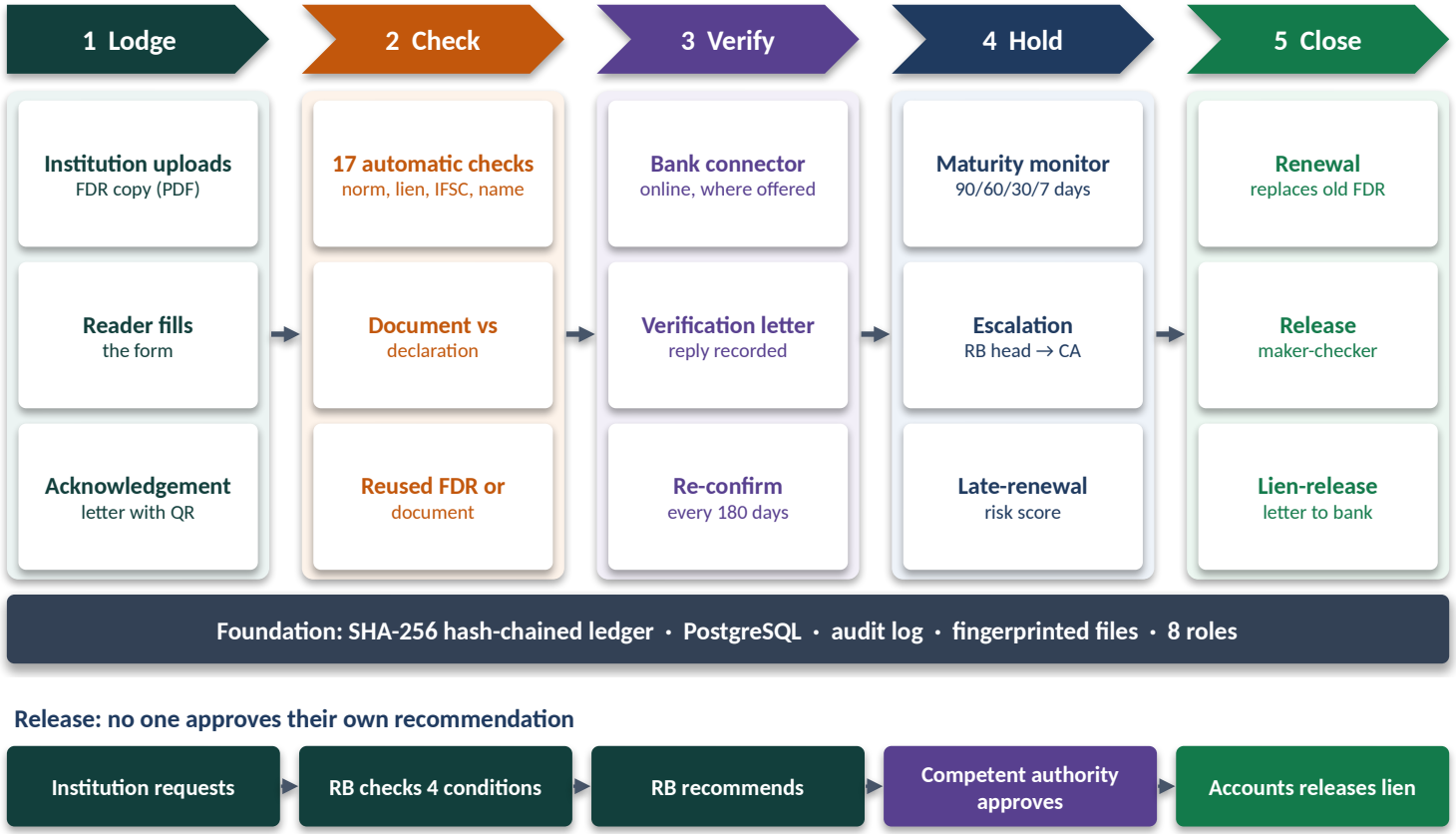
FULLY WORKING PROTOTYPE · all 6 PS test scenarios automated · 82 automated tests · 600 institutions, 1,849 deposits, 11 banks (all synthetic)

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TECHNICAL APPROACH

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Architecture: the life of a deposit



Release: no one approves their own recommendation

Every step is sealed in the deposit's own hash chain and in one system ledger; any later edit is detected and located.

Declared · document · bank, field by field

Field	Declared	FDR copy	Bank
Amount	₹5,00,000	₹50,000 ×	₹5,00,000 ✓
Depositor	Tapasya Edu. Society	same ✓	other trust ×
Lien	MS, AICTE	MS, AICTE ✓	MS, AICTE ✓

Banks with an online service answer through a connector (simulated in the demo); others get a letter and the officer records the reply.

Technology stack

Front end: React 19, TypeScript, Vite, Tailwind CSS 4, TanStack Query, Recharts; Hindi and English

Back end: Python, FastAPI, SQLAlchemy 2, Alembic; live updates by Server-Sent Events

Documents: pypdf and Tesseract OCR (reading FDR copies, scans, photos), ReportLab (letters, QR codes, PDF reports), openpyxl

AI/ML, all offline: scikit-learn gradient boosting, RapidFuzz name matching

Security: AES-256-GCM encryption of stored files, SHA-256 hash chain, bcrypt + JWT

Data and ops: PostgreSQL 16 or SQLite, Docker, Playwright browser tests

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FEASIBILITY AND VIABILITY

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1.00 / 1.00	precision / recall of the checks on 500 test deposits with planted faults; **0 of 50** clean ones flagged
100%	of fields read correctly from 300 test FDR copies in 3 bank layouts
81%	of the top 10% flagged by the late-renewal model renewed late (base rate 44%; AUC 0.65)
~1 s	to re-verify all 15,091 sealed ledger entries and 1,849 deposit histories

Feasibility

Technical

- Built and running; every number above is re-measured on the Proof page. Open-source stack on one server, NIC cloud or a state data centre.

Operational

- Institutions lodge and track their own deposits; officers stop retyping. Bank letters, rectification notices and lien releases are generated.

Economic

- No licence fees and no per-use AI charges; the bank connector is the only integration cost.

Legal and data

- **DPDP Act 2023:** data stays on AICTE's server; files encrypted at rest; institutions see only their own records; contact details masked for read-only roles; every export logged.

Challenges and how we handle them

Challenge	Mitigation built into Nidhi
No online FDR verification at many banks	Letter route with QR; officer records the reply; online connector where a bank offers one
Altered or reused FDR copies	Copy read and compared on lodging; FDR and file fingerprints checked across all institutions
FDR closed at the branch after acceptance	Every held FDR re-confirmed with the bank every 180 days
Renewals noticed too late	Reminders from 90 days; escalation at 15 and 45 days; likely-late list
Release without authority	Four conditions, maker-checker, approver ≠ recommender, sealed history
Scanned copies and phone photos	Tesseract OCR on the server: 99.6% of fields on 45 test scans; officer and bank remain the backstop

Viability and roll-out

Month 1–2 — Pilot with the Regulation Bureau; import the existing deposit register

Month 3–4 — Institutions lodge renewals online; letter-based verification with banks

Month 5–6 — Bank verification APIs where available; AICTE portal sign-on; SMS and e-mail

Beyond AICTE: any regulator or department that holds bank guarantees, earnest money or FDRs can use it; deposit norms, reminders and escalation are settings.

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IMPACT AND BENEFITS

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Who gains what

Regulation Bureau
One register; scrutiny desk with the bank's answer beside the declaration; letters written for them.

Competent authority
Releases arrive checked and recommended; escalations of lapsed FDRs reach them automatically.

Finance & Accounts
Executes only approved releases; the lien-release letter is generated and verifiable.

Institutions
Lodge online, see what is required, get reminders before maturity and every letter in one place.

Banks
Scan the QR to confirm a letter is genuine; answer verification requests online where they can.

Audit / vigilance
Every decision named, timed and sealed; tampering is detected and located.

What the PS asks to improve, and how it shows

Record accuracy	three sources compared field by field; 17 checks on every lodging
Verification time	online where banks offer it; letters chased after 7 days
Timely renewal	reminders, escalation, likely-late list; on-time rate on the dashboard
Manual effort	form filled from the FDR copy; letters and reports generated
Pending visibility	every pending item has a queue, an owner and an age

PS test scenarios, automated

✔️ **New FDR submission**
read, lodged, acknowledged, bank-confirmed, accepted

✔️ **Validity tracking**
active → maturing → matured, by the monitor

✔️ **Approaching maturity**
reminders at each offset, sent once

✔️ **Renewal**
old FDR closed as renewed; chain kept

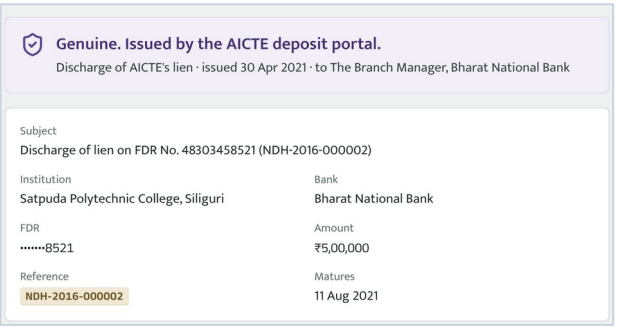
✔️ **Discrepancy**
short amount, no lien, altered copy, reused FDR, bank differs

✔️ **Authorised release**
checklist, maker-checker, lien letter verified by its PDF

✔️ **Also in a real browser**
17 Playwright tests drive the same flows end to end



Renewal history: an institution that renewed late three times; its next FDR is flagged early



Public letter check: a bank branch confirms a lien-release letter by its QR

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RESEARCH AND REFERENCES

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Today's way of working, and Nidhi

Need in PS 26265	Files, letters, spreadsheets	Nidhi
One up-to-date register of deposits	Partial, per desk	✔
Institution-declared vs bank-confirmed kept apart	✘	✔
Altered or reused FDR detected	By chance	✔
Reminders before maturity, escalation after	Manual	✔
Release with recorded conditions and approvals	On file notings	✔
Letters verifiable by the bank	✘	Y (QR)
Tamper-evident history	✘	✔

Comparison drawn from the problem statement's own description of current practice.

References

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https://www.iso.org/standard/62021.html

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[6] scikit-learn: HistGradientBoostingClassifier; RapidFuzz: string similarity
https://scikit-learn.org

Prototype and resources

Live prototype
[add link after deploying]

Demo video
[add link]

Source code
[add GitHub link]

Demo sign-in
rb.head@nidhi.test · kadamb@nidhi.test · password Demo@2026